

Policy vs. Public Opinion: An Analysis of Artificial Intelligence Trends

Group number 7

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Abstract

With the rise of various artificial intelligence technologies over the past decade, there has been a corresponding growth in both legislation and public opinion on the matter. To ensure the policy is supporting the needs of the public, this study examines the trends in topics between the two datasets and looks for potential misalignments of topic, tone, or sentiment. Looking first into the legislative side through a dataset created by a policy research group within Georgetown University, there is a clear increase in action. Next, looking into social media comments on the popular platform known as Reddit, the discourse about the topic has also grown. To examine the trends between the two, we utilized methods such as VADER sentiment analysis, BERTopic clustering, zero-shot text classification, and Detoxify tone analysis.

We compared the two domains using four methods: VADER sentiment analysis to monitor positive or negative sentiment on social media, BERTopic for topic clustering without supervision, zero-shot classification for assigning labels to particular issues (e.g., ethics, surveillance, misinformation), and Detoxify for monitoring for toxicity or emotionally laden content. We found that public and legislative concern with AI has grown, but with a diverging thematic focus. Both sides exhibit a consistent focus on AI bias and fairness, but the public concern with misinformation and risks of mass adoption far exceeds legislative concern with these issues, to the extent that they are infrequently highlighted in legislative proposals. We discovered that bursts of negative sentiment on Reddit predicted legislative activity and that public debate may serve as an indicator of the policy response.

These findings suggest a disconnect between issues that matter to the public and the priorities of policymakers in AI governance. Continuing to bridge these gaps will require quicker policy implementation that allows for responsiveness and clearer feedback loops between citizens and government legislators. We hope our findings lay a data-driven framework for improving

future legislation related to AI, and visibility for policymakers on the public concerns that matter.

Introduction

With the rapid advancement of artificial intelligence over the past decade, this technology has spread to every corner of society. According to a recent Pew Research study, experts estimate that nearly 80% of people in the U.S. interact with AI several times a day, with over 95% interacting at least once a day (McClain et al., 2025). It has taken over several spaces that regular individuals utilize daily, such as fitness, online commerce, and online support. This will only continue to grow as companies integrate AI further into their products and services. From education to finance to healthcare, AI is involved in decision-making, customer service, and policy. This influence has broad societal implications due to growing concern over safety, privacy, and misinformation. Legislators have been working to develop and implement policy in order to help protect against the potential harms of artificial intelligence. Regulations to safeguard both the public and private sectors seem evident, but it is also essential to address individual concerns over privacy. This work aims to address the overall alignment of AI policy goals and public concern by looking to answer the following questions:

RQ1: How have AI regulations and public opinions on AI evolved over the past decade?

RQ2: How has AI legislation been adopted in response to growing public concern?

In order to answer these questions, we created a multi-step pipeline to progressively deepen our understanding of both domains. To start with, we used EDA to discover the baseline trends of Reddit comments and documents. We considered activity over time, the metadata, and the terms used with this analysis. In short, we were able to address our first research question regarding the evolution of public opinions and subsequently legislation over the last decade. We then used BERTopic to extract dominant topics from the two datasets while not relying on predetermined terms. However, it did allow us to see that before we even mentioned legislation or policy, prominent themes often included surveillance, misinformation, and algorithmic fairness. We then used zero-shot classification because it allowed us to code each document to shared issue categories to assess how similarly public discussion and resulting policy match up, which we wanted to use as evidence for our second research question about the dynamism of legislation responding to public sentiment. Next, we used VADER sentiment analysis to assess whether and how themes were discussed emotionally by tracking the emotional valence over time. For example, we would have expected to identify whether spikes in positive or negative valence coincided with policy activity. Finally, we used Detoxify to identify in what moments Reddit discussions indicated heightened emotion, like nearly explosive frustration or outrage, when individuals might make decisions related to or act on political agency. Each of these stages resided on top of one another: EDA provided contextualization, BERTopic uncovered topics, zero-shot classification allowed for comparison, VADER tracked changes in sentiment, and Detoxify helped to discover moments of heightened emotions. Collectively, our use of methods provides a cohesive framework for thinking through how public concerns and AI policy co-evolve over time.

Related Works

The ethical, social, and regulatory implications of artificial intelligence have become a focal point of discussion. Increasingly, scholars have reported on public attitudes towards AI and the efforts of governments to oversee the development of AI. A 2025 Pew Research Center report reveals that Americans broadly identify AI as having a number of potential public benefits but equally express considerable concerns over issues like surveillance, misinformation, and bias [1]. Dreksler et al. at the Brookings Institution echo these concerns by arguing that the public has the right to expect AI systems to be transparent and accountable, and ultimately that those expectations must be more directly embedded into governance frameworks [2]. There has been considerable attention devoted to regulation, particularly relating to major initiatives, like the European Union AI Act, which seeks to provide an oversight regime for AI based on risk [4]. However, research comparing the U.S. to other settings by Podnar et al. (2025) found that regulation is even more modular and haphazard in the U.S., leading to what they call "significant misalignment" between policy frameworks and the pace of technological change [5]. The Stanford 2023 AI Index further identified that while there is rising public concern globally about AI, there were few effective mechanisms for incorporating these public sentiments into contemporary oversight evaluations, particularly in real-time [3]. While we have seen increasing work in both the public perception literature and policy development literature, few studies have attempted to examine the direct relationship between the two areas of study. Our study aims to close this gap by evaluating whether the public concerns expressed through policy and social media are actually being translated into the legislative process. Building on work across the fields of NLP, policy, and social media, we provide a new lens for understanding the responsiveness of AI oversight to public sentiment.

Data and Methods

This study was built around a pre-made AI Policy database discovered on Kaggle, created by the Emerging Technology Observatory of Georgetown University[9]. We focused on a singular file of AI documents that spans nearly a decade from 2016 to 2024 and contains over six hundred AI-related laws, policies, and standards covering countries around the world. Each data point contains:

- The full text of the published legislation
- A short and long summary
- Publication date and last edit date
- Issuing authority
- True/false flags for risk category, harm category, or other pre-assigned label

To complement this policy data with public sentiment, we collected Reddit comments from AI-focused subreddits including r/MachineLearning, r/ArtificialIntelligence, and r/OpenAI using the PRAW Python API. We queried posts and top-level comments containing artificial intelligence keywords and filtered them to the same time frame as the AGORA policy documents (2016-2024). Well over two thousand comments were pulled, each containing the following fields:

- Post ID

- Title
- Comment text
- Timestamp
- Subreddit
- Upvote score

This cleaned Reddit set, aligned time-wise and structurally with the AGORA policy data, enabled the next steps of side-by-side analyses of tone, topic, and toxicity. The table below summarizes the two datasets.

	Number of observations	Avg. Length	Period	Number of Features	Key Fields
AI policy docs	402	850 words	2016-2024	100	Date, summary
Reddit Comments	2352	40 words	2016-2024	8	Timestamp, comment text

Figure 1: Dataset Descriptive Statistics

From here, we dove into a complex and varied analysis involving several distinct methodologies. A wide range of methods were used, but only four provided applicable insights to this study. The first of these methods is VADER sentiment analysis, which we used on both the Legislative and Reddit datasets. VADER is a pre-trained natural language processing model specifically designed for social media analysis. The model utilizes its pre-trained dictionary of words along with punctuation and capitalization to assign each datapoint an average positive, neutral, and negative value as well as an overall composite score from -1 to +1. We choose to aggregate annually to provide for a better visualization of the trends, to demonstrate a side-by-side analysis, and to answer our question of how AI evolves.

Then, we investigated the dominant topics in each dataset using BERTopic to help us better understand the common driving issues in discourse. Prior to modeling, we pre-processed the data, removing stop words, punctuation, and other noise, and routed the cleaned text through the BERTopic pipeline. We played with parameter tuning, and once we generated the topic groupings, we sorted them by frequency and reviewed the key terms and representative documents in each cluster to help interpret and ascribe meaning to the clusters. Next, we chose the top 10–15 most coherent and frequent topics from each dataset and compared how certain issues were highlighted in public discourse versus legislative texts. We also tracked topic frequency across time to consider whether attention to issues similarly surged or waned in both contexts.

Next, we applied zero-shot text classification to both of the datasets to sort each document into one of the four main AI theme categories: “Privacy”, “Bias”, “Transparency”, and “Misinformation”. After minimal processing, we loaded the zero-shot classification pipeline using the Facebook Bart-large-ml

model. Due to the size of the Reddit data, we had to select a random sample of about 10% of the dataset due to computational complexity, since it took about ten minutes per hundred observations. We selected a random sample of 250 observations from both the Reddit and AGORA datasets. The results produced are visible in the side-by-side figure below. For each dataset, we captured the single label assigned by the classifier and tallied the results for comparison, which is seen in the next section.

count		count	
predicted_label		predicted_label	
Bias	92	Bias	92
Transparency	86	Misinformation	91
Privacy	53	Transparency	44
Misinformation	19	Privacy	23
dtype: int64		dtype: int64	

Figure 2: AGORA (left) versus Reddit (right) text classification

Finally, we utilized Detoxify on the Reddit dataset to measure the frequency and intensity of toxic language throughout time. After conducting the necessary text cleaning/formatting, we passed the individual-level Reddit comments through the model, treating each entry as a separate observation. The model generated a toxicity score for each of the comments, anywhere between zero and one, so we averaged the scores each year so that we could flag shifts in intensity in rhetorical patterns. This approach helped us to better assess public sentiment through emotional intensity and how that might function as an indicator of public policy action.

comment_text	comment_score	comment_created_utc	toxicity	severe_toxicity	obscene	threat	insult	identity_attack
The next big thing will be something no one us talking about yet. The next pretty big thing, though, will be in biotech. CRISPR, biofuels, new bio-based degradables.	4	2018-06-19 05:06:46	0.0037557688	0.00013066747	0.00032609768	0.00018227377	0.0003346373	0.00031817696
Machine learning is at the top of the list IMO with robotics a close second. I would throw in two others. Battery Technology and real-time Holography.	2	2018-06-19 05:02:18	0.00068634085	0.000114098344	0.00017342735	0.00012090058	0.00018138542	0.0001413496

Figure 3: Sample data after applying the Detoxify model to Reddit comments

Results

Beginning with the results from the VADER sentiment analysis, we see below in Figure 4, the circle size shows the document count each year as a reminder that years with fewer text will look a bit jumpy due to fewer observations.

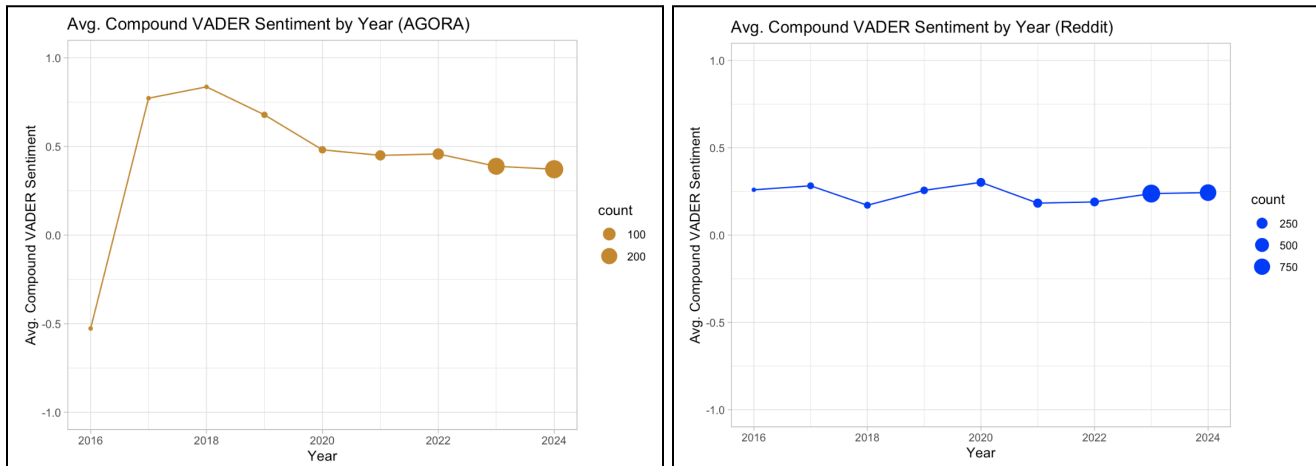


Figure 4: Average Compound VADER Sentiment

Examining the curves, Reddit comments remained within a mild-positive range from 2016 to 2024, with dips in sentiment observed in 2018 and 2021. These dips seem to coincide with major AI milestones, including the EU's GDPR of 2018 and OpenAI's release of GPT-3[7, 8]. On the other side, the AGORA composite scores start quite negative and then shift to very positive until settling in a similar mild-positive zone as the Reddit values. This inconsistency suggests that legislative tone may initially lag behind public excitement but ultimately stabilizes. We choose to further explore that lag in policy reaction with another method below.

Next, our BERTopic analysis revealed notable thematic patterns. We found some overlapping topics between the two datasets, such as a heavy focus on misinformation, but there were apparent differences in some of the topics. AI laws were more focused on national security and economic development issues, while public discourse was much more concerned with areas connected to immediate social changes, like job concerns. These differences emphasized a substantial discrepancy between public opinion and legislation. The public concerns of AI, especially in areas like security and misinformation, may not be getting fully addressed in the current and past legislative frameworks, possibly hinting at the need for reform to better inform and reflect the public.

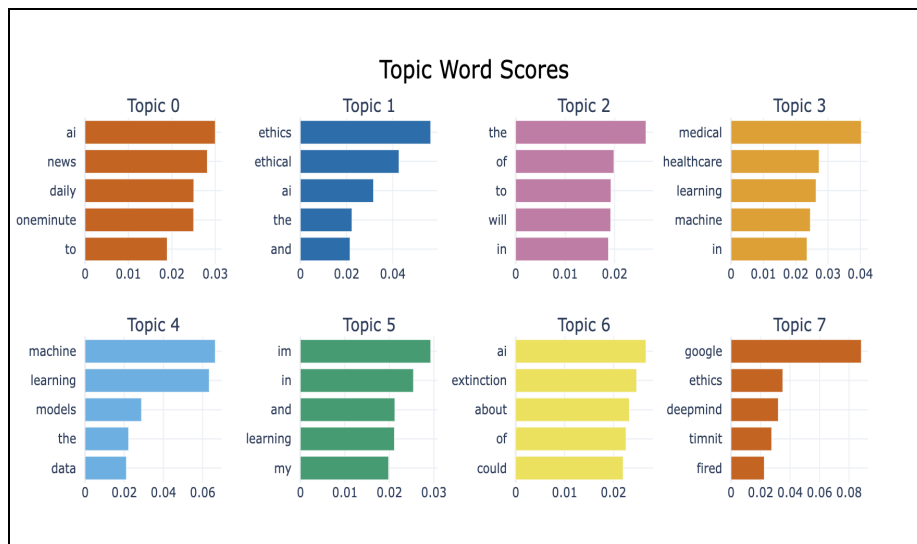


Figure 5: BERTopic classifications

Moving to zero-shot text classification, we worked to examine how these topic trends varied between the legislative and Reddit datasets. In the AGORA legislative set, the prioritization order was Bias, Transparency, Privacy, and Misinformation last, with only 7.6% of documents falling into this category. For the Reddit sample, the prioritization order also led with Bias, but Misinformation was a close second, followed by Transparency and Privacy. These counts suggest that both policy and public opinion share concern about Bias; however, regarding their second priorities, opinions differ. This is further visualized in the figure below, where the contrast is quite apparent.

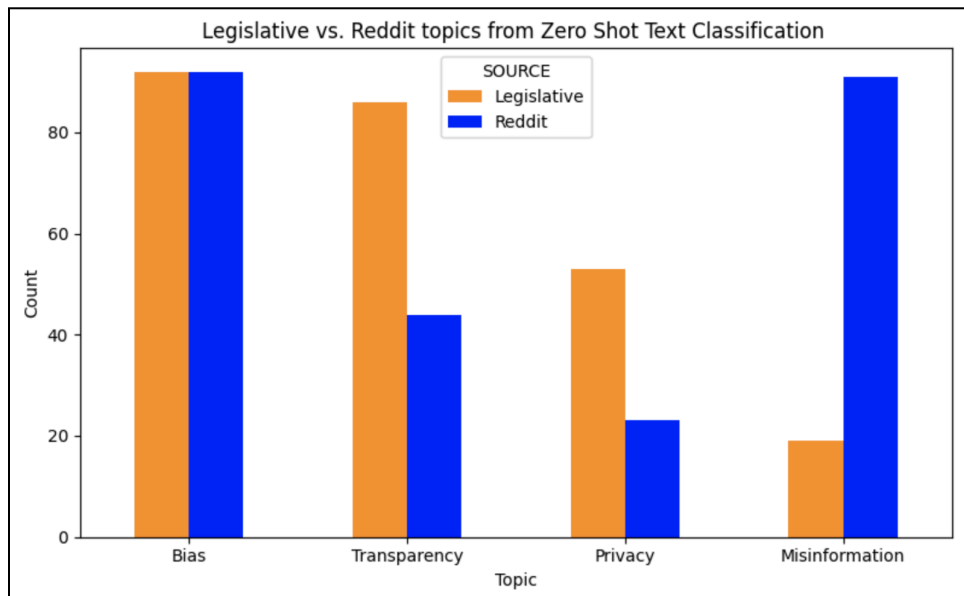


Figure 6: Zero Shot Text Classification category comparison

Finally, the Detoxify model allowed us to examine the emotional intensity of Reddit conversations. As shown in Figure 7, we found spikes in toxicity corresponding with high-profile AI controversies or

releases. Similar to the patterns we saw with the sentiment analysis, we saw toxicity scores increase in 2020 and 2023. After some research on AI events, these increases seem to align with events like public debate on facial recognition and open-access LLMs and Chat model concerns[10, 11]. These tone shifts in toxicity serve as indicators of negative public sentiment, meaning this type of tool could be beneficial in tracking real-time public opinion.

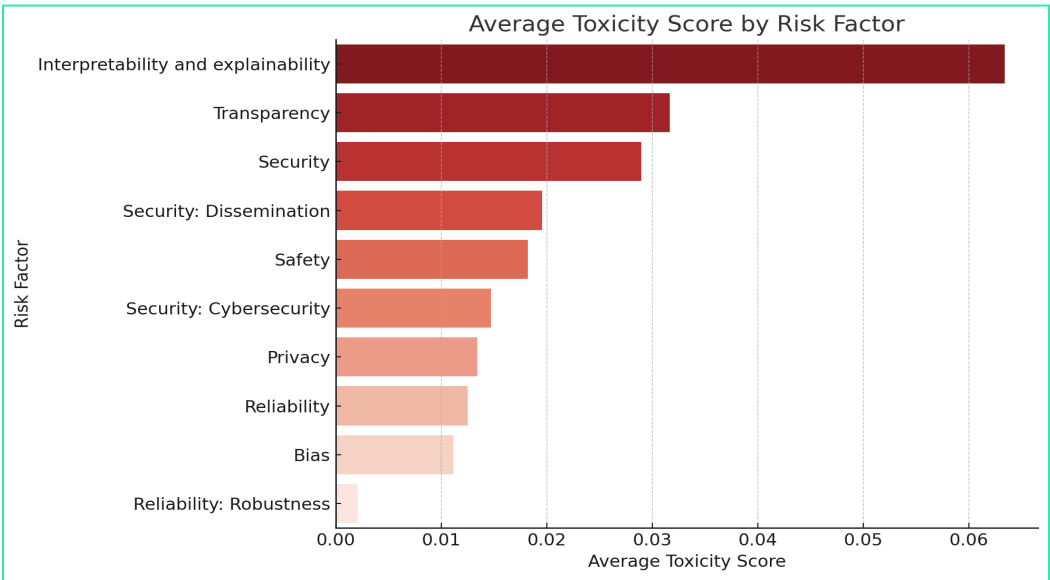


Figure 7: Average Toxicity Score by pre-assigned risk factor

Discussion

Our findings show that while both public conversation and policy attention around AI have grown over the past decade, the two are not always aligned in terms of sentiment. While both seem to have moderate concern over AI bias, their focuses in other categories differ dramatically. This difference suggests that even though policy is progressing, it might be missing some of the issues that matter most to the public. One interesting pattern we noticed is that while sentiment on Reddit stayed relatively steady over time, the tone of policy documents shifted more dramatically from negative to highly positive before settling somewhere in the middle. That could mean policy responses are reactive, catching up only after public sentiment peaks. We also found that spikes in negativity or toxicity on Reddit often came just before major policy activity, which makes us think public discourse might actually be a useful early signal for when legislation is needed. For now, most governance structures don't seem built to pick up on or respond to this real-time feedback, but they probably could if implemented into the feedback. If public concerns like misinformation or mass adoption risks are overlooked for too long, it could undermine trust in both AI and the institutions or personnel that oversee it. Based on our findings, we think policymakers would benefit from engaging more directly with the tone and themes of public discussion, not just technical input from experts. A faster, more responsive feedback loop between citizens and lawmakers could help AI policy stay better aligned with what people actually care about.

Limitations and Future Work

While this study has provided useful insight into the relationship between public sentiment and legislative action on artificial intelligence, there are several limitations that should be considered. First, the Reddit dataset is not representative of public sentiment as a whole, as it tends to be biased toward younger, technology-oriented users and is not inclusive of other online forums such as Twitter or Facebook, nor traditional survey data. We also did not consider the upvotes or downvotes of each comment, which could hold significant weight in their representation of public opinion. This is something that could be done in future work using these methods. Further, even though the VADER sentiment analysis tool has been shown to perform well for short, informal text, it will miss subtle differences in emotion, figure, or sarcasm reflecting a nuanced position, typical in online discourse. It is also meant for social media analysis, not legislation, indicating our comparison of the model on the two datasets may not be entirely logical. Finally, we attempted to use zero-shot classifiers, while useful, but were limited by the quality of the label and training set data associated with the pre-trained model. These data may not fully represent the constantly shifting contours of issue categories. Also, while we were able to conduct descriptive and correlation analysis due to time constraints, we were not able to consider the extent to which public sentiment influences a change in policy or the introduction of new policy. Future research to address these limitations could be studied with a larger set of public discourse platforms, as well as newer forms of more robust sentiment models, such as transformer-based classifiers. Lastly, developing interactive dashboards or real-time monitoring of public sentiment pertaining to emergent AI issues could help to improve how policymakers track evolving public sentiment and, conversely, what implications these public needs could have for governance.

References

- [1] Pew Research Center. How the US Public and AI Experts View Artificial Intelligence. Washington, D.C.: Pew Research Center, 2025.
- [2] Dreksler, Noah, Sophie Clare, Kenneth J. Schiff, and Darrell M. West. What the Public Thinks About AI and the Implications for Governance. Washington, D.C.: Brookings Institution, 2025.
- [3] Stanford Institute for Human-Centered Artificial Intelligence. The AI Index Report 2023: Chapter 8 – Public Opinion. Stanford, CA: Stanford University, 2023.
- [4] European Commission. The EU AI Act – A Harmonised Regulatory Framework on Artificial Intelligence. Brussels: European Commission, 2024.
- [5] Podnar, Sofia, Gianna Marolda, and Simon Lund. “Barreling Towards ‘Significant Misalignment’: A Comparative Examination of AI Regulation in the European Union and United States.” *Pittsburgh Undergraduate Review* 4, no. 1 (2025): 44–61.
- [6] Turner Lee, Nicol. Testimony: How the White House Executive Order on AI Ensures an Effective Governance Regime. Washington, D.C.: Brookings Institution, 2024.
- [7] “What Is GDPR, the EU’s New Data Protection Law?” *GDPR.eu*, August 29, 2024.

[8] Heaven, Will Douglas. “2021 Was the Year of Monster Ai Models.” MIT Technology Review, May 11, 2022.

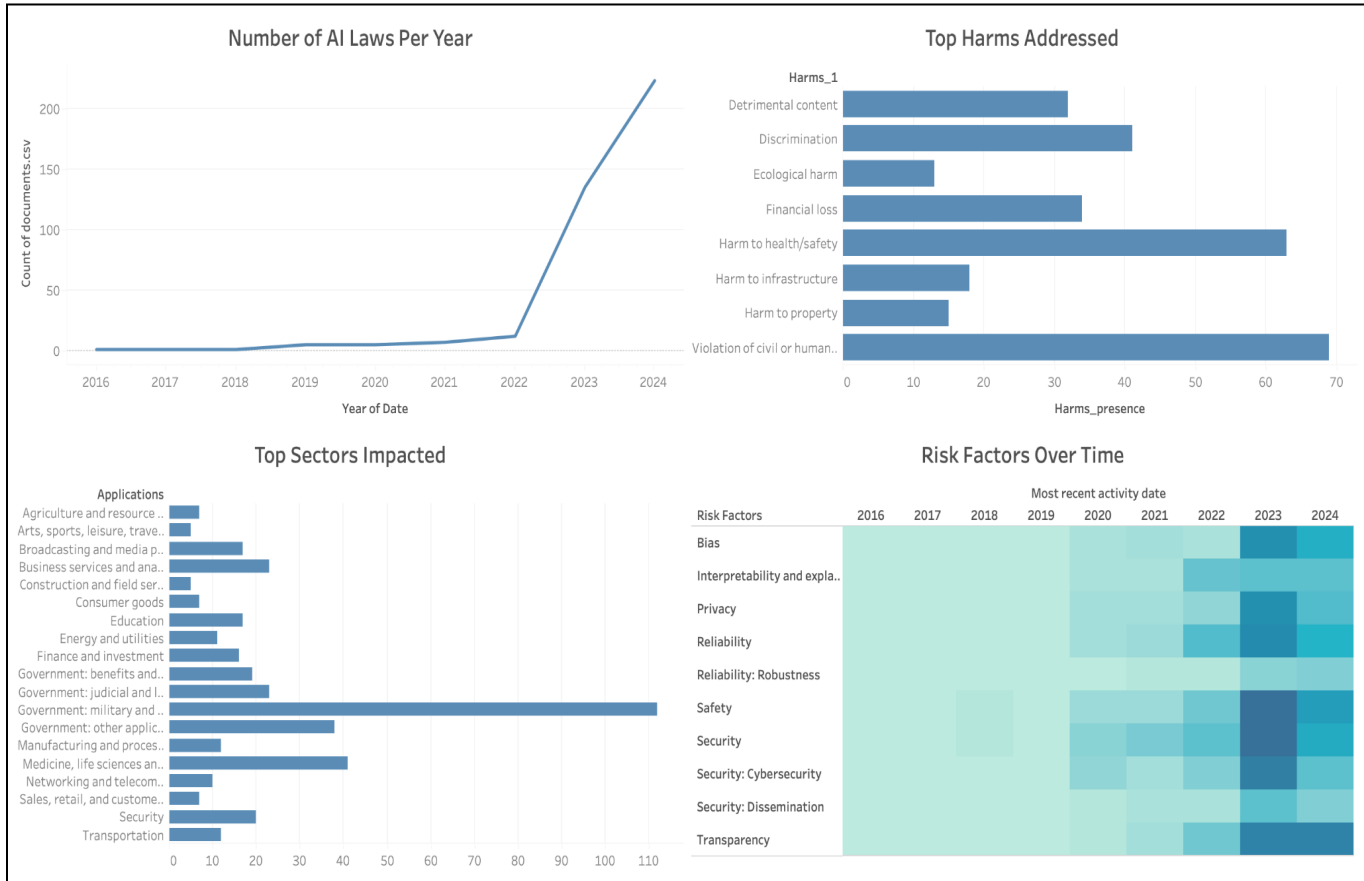
[9] Haddii, Umer. “AI Governance Documents Data.” Kaggle, January 12, 2025.

[10] Wrongfully accused by an algorithm - The New York Times. Accessed June 14, 2025.

[11] “Large Language Models Pose Risk to Science with False Answers, Says.” University of Oxford. Accessed June 14, 2025.

Appendices (Optional)

[1]



More resources utilized but not cited in report:

<https://huggingface.co/facebook/bart-large-mnli>

<https://github.com/mathisonian/awesome-visualization-research?tab=readme-ov-file#graphs>

<https://www.geeksforgeeks.org/graph-plotting-in-python-set-1/>

<https://www.theguardian.com/technology/2016/dec/28/2016-the-year-ai-came-of-age>